

- 4 In the immune system, a plasma cell develops from an activated B-lymphocyte. Mature plasma cells synthesise and secrete antibody molecules.

(a) Fig. 4.1 is a diagram of a transmission electron micrograph of a plasma cell.

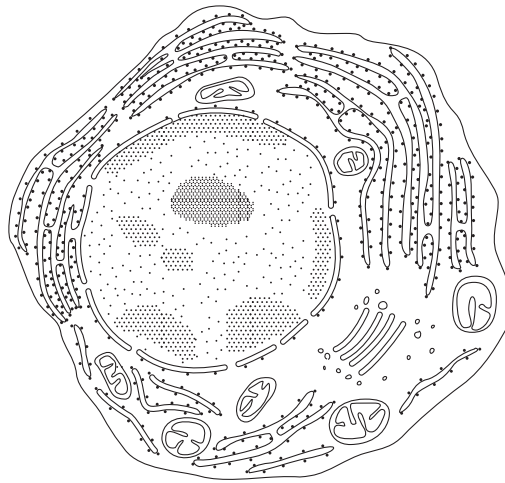


Fig. 4.1

The plasma cell can be seen in greater detail using an electron microscope compared with using a light microscope.

- (i) Describe the **extra** detail of the nucleus that can be seen using an electron microscope.

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- (ii) Explain why cell structures, such as ribosomes and the rough and smooth endoplasmic reticulum, cannot be seen using a light microscope.

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- (b) The transition from the activated B-lymphocyte to the fully mature plasma cell requires a number of mitotic cell cycles to occur. This process, which is known as clonal expansion, results in a large number of genetically identical plasma cells.

Fig. 4.2 describes events, **A** to **F**, that occur during the mitotic cell cycle of the B-lymphocyte.

- A** centrioles replicate
- B** DNA polymerase catalyses the formation of phosphodiester bonds
- C** condensation of chromosomes
- D** nuclear envelope reassembles around each set of daughter chromosomes
- E** centromeres move towards poles
- F** chromosomes line up at spindle equator

**Fig. 4.2**

Table 4.1 lists the stages occurring during one cell cycle of the B-lymphocyte. These stages are not in the correct order.

**Table 4.1**

| stage of cell cycle  | correct letter from Fig. 4.2 |
|----------------------|------------------------------|
| G <sub>2</sub> phase |                              |
| metaphase            | <b>F</b>                     |
| cytokinesis          |                              |
| prophase             |                              |
| S phase              |                              |
| anaphase             |                              |
| G <sub>1</sub> phase |                              |
| telophase            |                              |

Complete Table 4.1 by writing the letter of the event described in Fig. 4.2 that correctly matches the stage of the cell cycle listed.

Leave a **blank space** if there is **no** matching description for the stage in the list. Use each letter **once** only.

One of the letters in Table 4.1 has already been added for you.

- (c) Clonal expansion also results in the production of memory B-lymphocytes.

Explain the importance of clonal expansion **and** the production of memory B-lymphocytes in providing protection for a person against an infectious disease.

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- (d) Myasthenia gravis is an example of a disease where the immune system fails to distinguish between self and non-self.

Explain what is meant by this statement.

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[Total: 15]